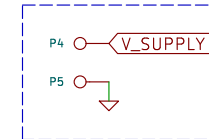
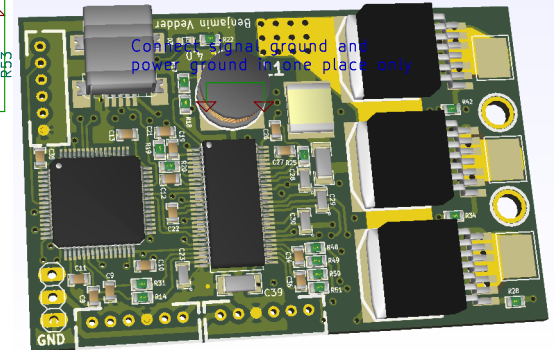
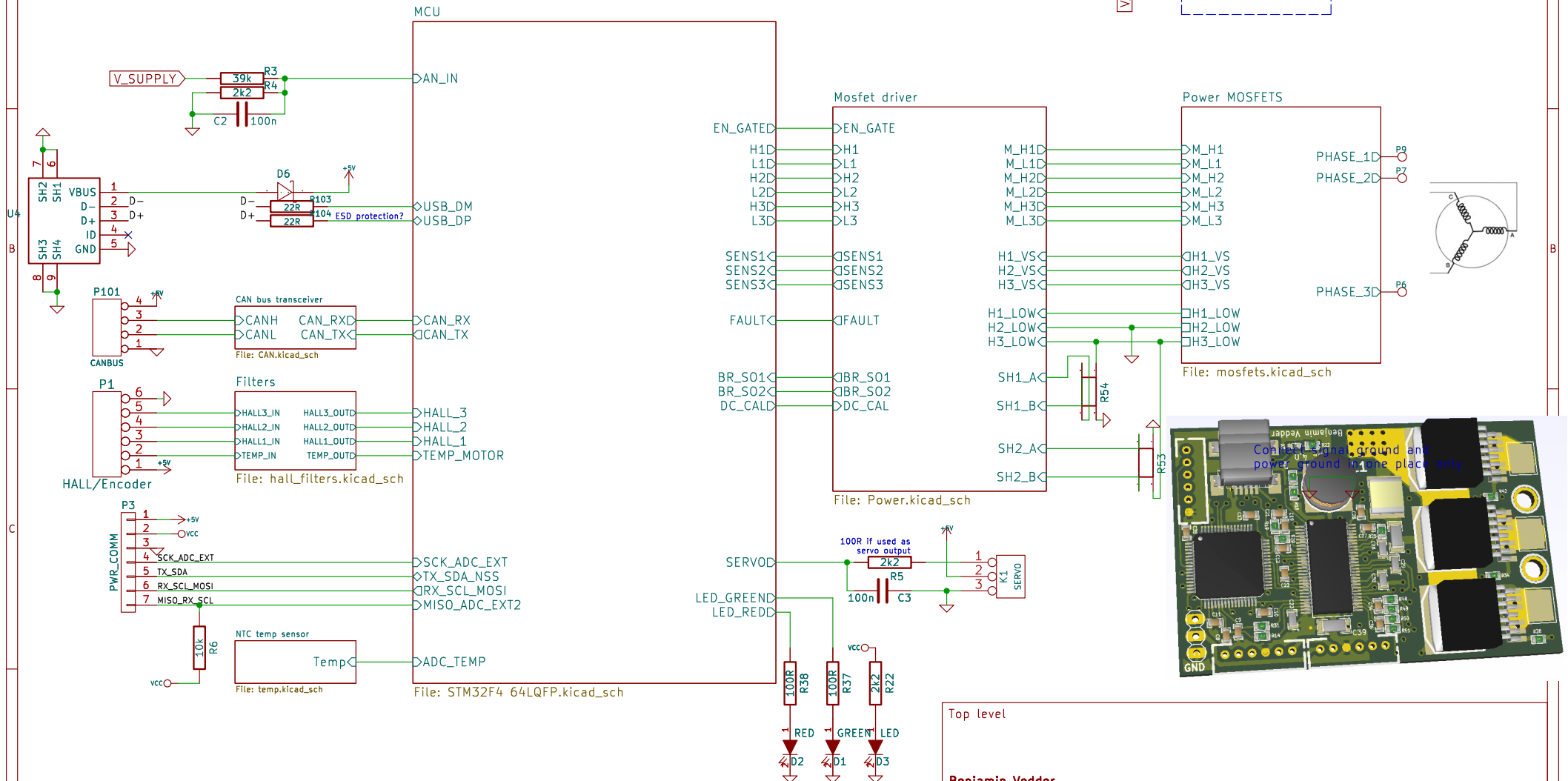


BLDC motor controller

H5
V_SUPPLY



Voltage supply (0 - 60v)
Needs external decoupling
caps to avoid high voltage
transients produced by the
inductance of the battery
wiring while switching the FETs
Also critical for EMI/RF compliance



Top level

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Sheet: /

File: after-remap.kicad_sch

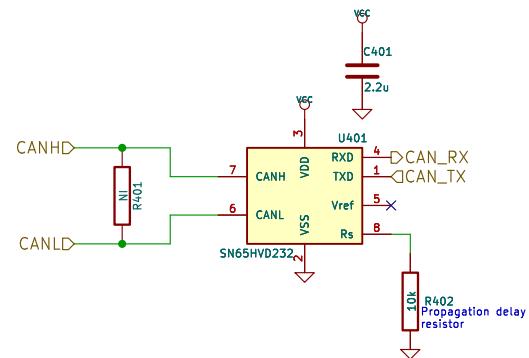
Title: BLDC Driver 4.11

Size: A4 Date: 21 aug 2015

KiCad E.D.A. 9.0.2

Rev: 4.12

Id: 1/7



Package Types

MCP2561 PDIP, SOIC 	MCP2562 PDIP, SOIC
MCP2561 3x3 DFN* 	MCP2562 3x3 DFN*

* Includes Exposed Thermal Pad (EP); see [Table 1-2](#).

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Sheet: /CAN bus transceiver/

File: CAN.kicad_sch

Title: BLDC Driver 4.11

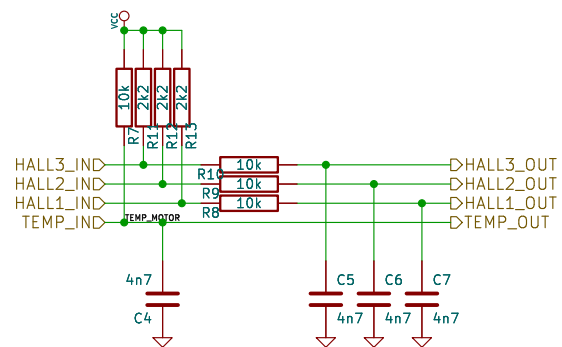
Size: A4

Date: 21 aug 2015

Rev: 4.12

KiCad E.D.A. 9.0.2

Id: 2/7



Benjamin Vedder

Sheet: /Filters/

File: hall_filters.kicad_sch

Title: BLDC Driver 4.11

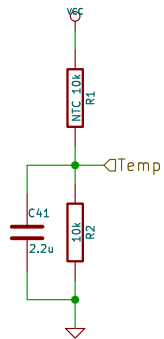
Size: A4

Date: 21 aug 2015

Rev: 4.12

KiCad E.D.A. 9.0.2

Id: 3/7



Benjamin Vedder

Sheet: /NTC temp sensor/
File: temp.kicad_sch

Title: BLDC Driver 4.11

Size: A4

Date: 21 aug 2015

Rev: 4.12

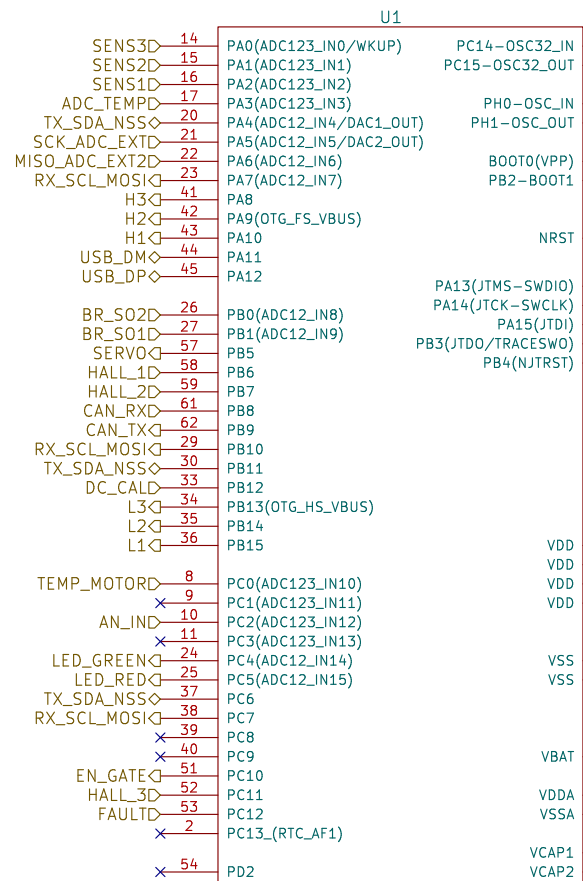
KiCad E.D.A. 9.0.2

Id: 4/7

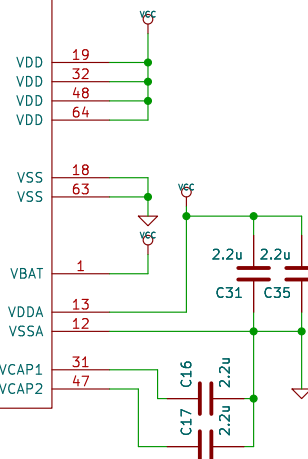
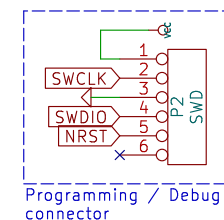
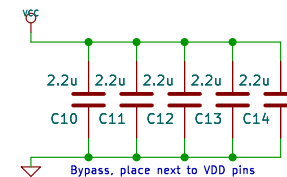
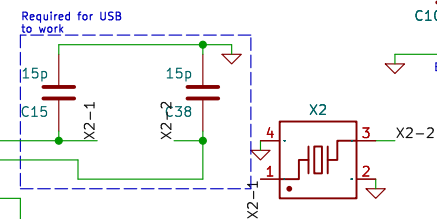


STM32F405xx STM32F407xx

ARM Cortex-M4 32b MCU+FPU, 210DMIPS, up to 1MB Flash/192+4KB RAM, USB
OTG HS/FS, Ethernet, 17 TIMs, 3 ADCs, 15 comm. interfaces & camera



STM32F40X_LQFP64



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Sheet: /MCU/
File: STM32F4_64LQFP.kicad_sch

Title: BLDC Driver 4.11

Size: A4 Date: 21 aug 2015

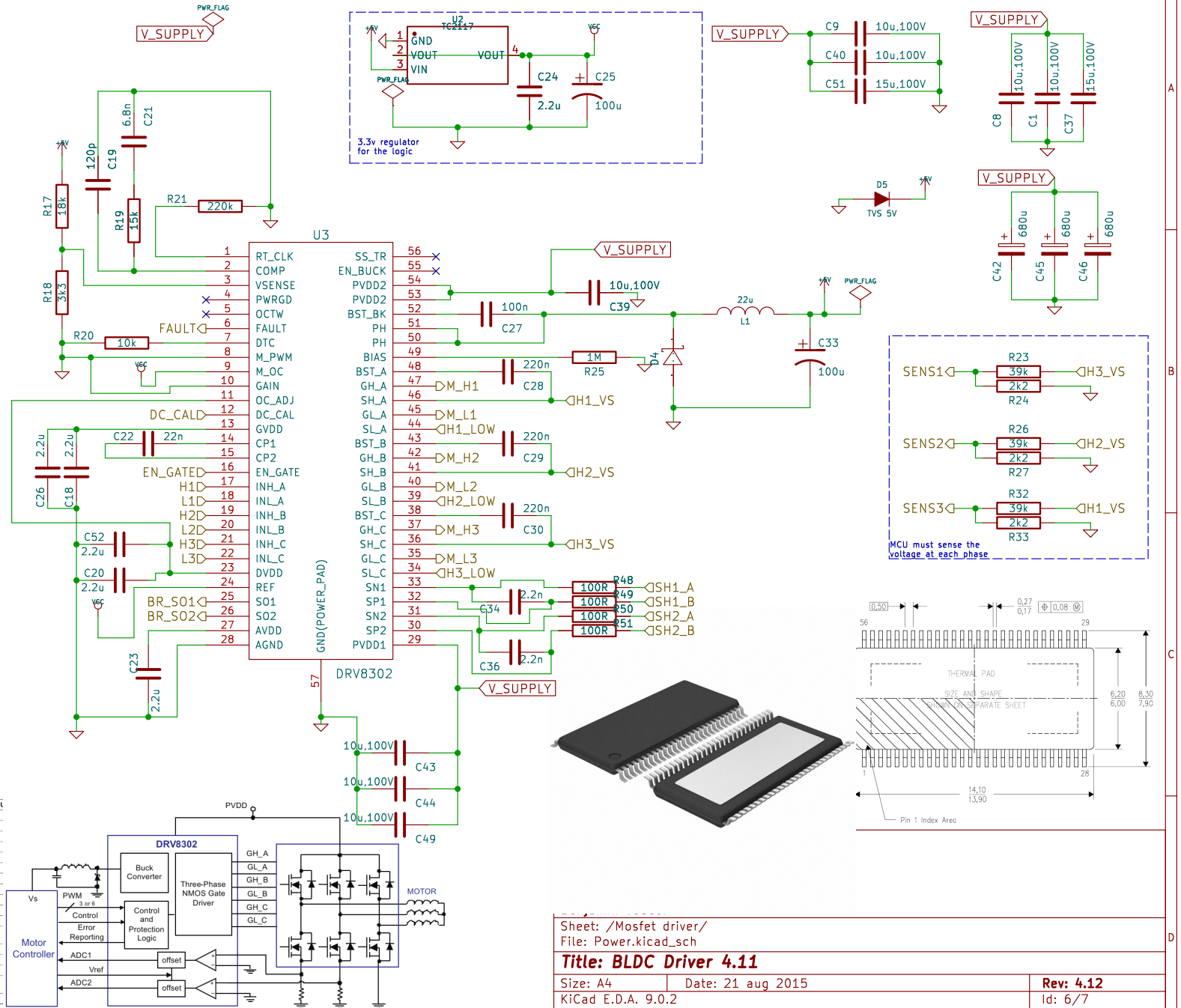
KiCad E.D.A. 9.0.2

Rev: 4.12

Id: 5/7

FEATURES

- Operating Supply Voltage 8V–60V
- 2.3A Sink and 1.7A Source Gate Drive Current Capability
- Integrated Dual Shunt Current Amplifiers With Adjustable Gain and Offset
- Integrated Buck Converter to Support up to 1.5A External Load
- Independent Control of 3 or 6 PWM Inputs
- Bootstrap Gate Driver With 100% Duty Cycle Support
- Programmable Dead Time to Protect External FETs from Shoot Through
- Programmable Overcurrent Protection of External MOSFETs
- Thermally Enhanced 56-Pin TSSOP Pad Down DCA Package

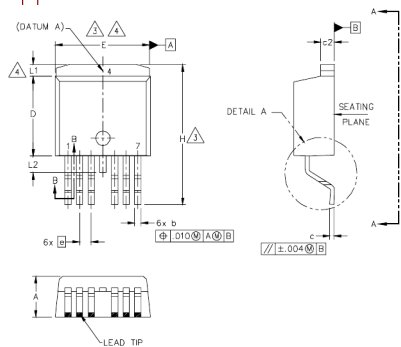
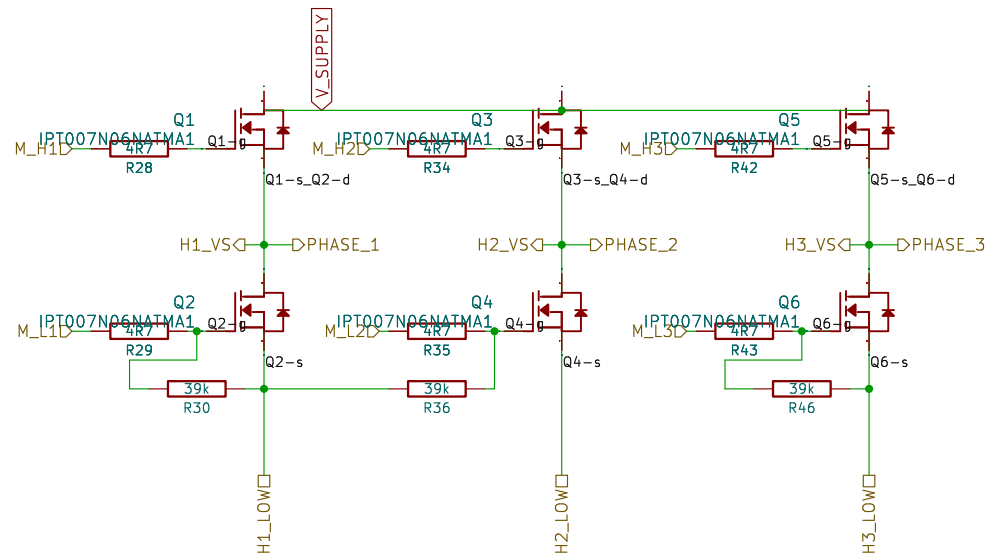


Sheet: /Mosfet driver/
File: Power.kicad_sch

Title: BLDC Driver 4.11

Size: A4 Date: 21 aug 2015
KiCad E.D.A. 9.0.2

Rev: 4.12
Id: 6/7



Benjamin Vedder

Sheet: /Power MOSFETS/

File: mosfets.kicad_sch

Title: BLDC Driver 4.11

Size: A4

Date: 21 aug 2015

Rev: 4.12

KiCad E.D.A. 9.0.2

Id: 7/7